Template coverage: legacy report templates in the V4 block schema

Status 30 October 2026, fourth version: classes follow table 3.3 of the shared document of 30 October (questionnaires are a later module; the status "to confirm" is gone; "Core, proposed" is new). Mockup analysis, pseudonymised. The legacy (V3) templates, their images and their texts are not in this repository (CLAUDE.md rule 2); clinics appear as Clinic A to H, templates as T01 to T16. Per-template models: `coverage/models/Tnn.json`; all tables below: `python3 -m mockup.coverage_summary`.

Answer in brief

Does the block schema of I6 cover about 90% of what the templates need?

- **For the first release (English and German templates): 83% as I6 describes the schema today, about 94% with three small additions,** and 100% with eight. The three that matter most: a device block for recording and norm facts (TG10), a clinician form block (TG6), and a generated table of contents (TG5).
- **For the Japanese templates (a later market): 62% today.** They need two more additions: neutral legend variants inside device figures (TG1) and Japanese plus English titles (TG2). Together with TG10 they reach 100%. These three templates carry 7,410 of the 9,638 reports a year in the sample, so the weighted figure over all templates is only 67%.
- **What can be carried over unchanged is much less: 68% of the English and German elements in scope, 50% of the Japanese.** The rest either needs a schema addition or holds claim wording: 70 of 153 clinic texts (46%) name conditions or mental functions, give treatment advice, or state a fixed verdict.
- **On top of that, 236 elements (22% of all) must be left out**, mostly fixed verdicts (79) and treatment advice (66). **349 elements (32%) belong to analyses outside the core**: 242 return when their modules are certified, 122 of them questionnaires and cognitive tests (table 3.3: a later module; until it is certified, clinics use questionnaires in V3), 9 the V3 sLORETA block, which localises selected ICA components; 75 never return (dropped or research only); and 32 depend on open points: ICA component ERPs and the ERPC2 checklist over them (22, Q7), microstates (7, table 3.3 Open, a question to the clinical lead), a lateral F7/F8 variant of the asymmetry index that table 3.3 does not list, and two text markers of unclear meaning (3).
- **Changes of 30 October:** the 19 elements "to confirm" (frequency markers 10, global coherence 9) are plain Core now and were already counted as covered; the 9 elements with V3's lagged, instantaneous and total global coherence move from Open to Core, and the V3 workload index (T10) from later module to "Core, proposed", so the elements in scope rise from 490 to
- Microstates move from later module to Open, the V3 sLORETA block from Open to later module. **The one "Core, proposed" element** (the frontal theta / parietal alpha ratio) is counted as covered here, but is covered only if the clinical lead agrees (table 3.3); its V3 wording is not carried over either way.

So the product owner's expectation of "about 90%" (I6) holds for the structure of the first release after three additions. It does not hold for content: about a third of what clinics put into their templates today cannot go into a core report.

Limits of these figures. They count template elements, a proxy for "needs", and every element counts once whatever its size. The element "Core, proposed" (table 3.3) counts as covered; without it the figures do not change at the shown precision. The 16 templates are 69% of the 13,946 template uses of the year to 27 October 2026, out of about 121 templates used (I11). The largest unmodelled share, 14.7%, is a file missing on the server, last used in April 2026, with the same file name as T03 (probably an earlier version; its language is taken from the file name). The CP templates in the sample ran the AI interpretation only 28 times in a year, against about 2,150 CP reports in all. The models are an agent reading of the templates and the legacy audit; neither the CTO nor the clinical lead has checked them.

Method

Each template is split into elements: one per figure, table, generated text, static text, image, slot group, header, footer and clinician area. Each element has one class.

Class	Meaning	In the coverage figure
device	a result of a "Core" row of table 3.3	covered
core_proposed	a result of a "Core, proposed" row (beyond the clinical lead's list of 5 October; needs his agreement)	covered
clinic	the clinic's own neutral content (text, image, slots, header and footer)	covered
clinician	filled per report by the clinician, including the signature at issue	covered
not_expressible	needed in a core report but needs a schema change (template gap TG1 to TG11)	not covered
leave_out	must not be carried over and needs no schema support	not counted
later_module, research_only, dropped, open	outside the core (table 3.3)	not counted

Coverage = (in scope minus not expressible) / in scope; weighted by reports a year (rounded to tens). "Carried over as is" also takes out elements with claim wording. Every element needing a schema change counts as not expressible, even when most of it is a device figure.

The models were made in two rounds by agents working from the original files: a first round by three agents, reviewed by seven independent reviewers (111 findings, 10 blocking: a missed cover page in six templates, an inconsistent metric, classes differing between agents), and a second round with one rule set for all 16 templates, a new extraction that reads content controls and text boxes, and a written answer to every finding. The second round was checked by script rather than by a second review panel, to save tokens: valid structure and counts, same class for the same marker across templates, no clinic, person or place names, no 4-word run of template prose and no 6-character Japanese run (the only matches are measure names and table 3.3 rows), no exact colours, no web addresses.

Per template

Template	Clinic	Kind	Lang	Reports a year	Elements	Device	Core, proposed	Clinic	Clinician	Not expressible	Leave out	Outside the core	Coverage	Carried over as is	Clinic texts with claim wording
T01	Clinic A	RG	JA	3,760	18	6	0	4	0	6	2	0	62%	50%	1 of 1
T02	Clinic A	RG	JA	3,240	18	6	0	4	0	6	2	0	62%	50%	1 of 1
T03	Clinic A	RG	JA	410	18	6	0	4	0	6	2	0	62%	50%	1 of 1
T04	Clinic B	CP	DE	550	84	11	0	15	0	7	19	32	79%	55%	8 of 13
T05	Clinic B	CP	DE	460	47	2	0	10	4	1	10	20	94%	88%	1 of 6
T06	Clinic C	RG	EN	280	92	16	0	21	2	7	20	26	85%	72%	6 of 15
T07	Clinic D	RG	EN	230	93	18	0	19	2	10	22	22	80%	67%	6 of 15
T08	Company standard	RG	DE	210	56	6	0	21	0	12	8	9	69%	56%	5 of 14
T09	Clinic E	CP	EN	120	102	11	0	18	1	6	24	42	83%	67%	6 of 13

Template	Clinic	Kind	Lang	Reports a year	Elements	Device	Core, proposed	Clinic	Clinician	Not expressible	Leave out	Outside the core	Coverage	Carried over as is	Clinic texts with claim wording
T10	Clinic F	CP	DE	120	116	13	1	21	2	8	26	45	82%	64%	7 of 16
T11	Clinic F	CP	DE	90	108	11	0	20	2	7	25	43	82%	65%	7 of 15
T12	Clinic G	RG	DE	90	101	15	0	21	6	4	24	31	91%	74%	8 of 16
T13	Clinic H	CP	EN	70	110	18	0	18	2	13	25	34	75%	63%	6 of 14
T14	Company standard	CP	EN	6	101	18	0	14	1	8	24	36	80%	66%	6 of 10
T15	Clinic F	CP	EN	1	13	0	0	7	1	1	2	2	89%	78%	1 of 3
T16	Company standard	RG	EN	1	8	0	0	0	0	0	1	7	none in scope		

Kind: RG is a classic report template, CP one with slots for the AI interpretation. T01 to T03 have the same content. T09 is an English copy of the T10 family, T11 is T10 without three sections, and T15 (a stand-alone nutrition recommendation) is from the same clinic. T16 (heart rate variability) lies wholly outside the core. T04 and T05 are used as a pair (appendix and summary). Reports a year are rounded to tens.

Schema additions, ordered by effect on the first release

#	Gap	Templates (reports a year)	Proposed addition	Weighted coverage EN and DE after adding it and those above
TG10	Recording parameters, amplifier, filters, montage, band legends and norm description typed as clinic text or drawn into clinic images; in T01 to T03 a cover figure shows band limits that differ from the device bands	T01 to T04, T06 to T14 (9,180)	Device metadata placeholders and a device method block; validation that clinic text does not state device parameters	87%
TG6	Forms the clinician fills: tick-box checklists, screening and findings tables, recommendation tables	T04, T06, T07, T09 to T11, T13, T14 (1,470)	A clinician form block with clinic-defined rows; rows must not name conditions, mental functions or treatments (15 of these forms do today); the clinical lead decides	91%
TG5	Table of contents and "see page N"	T04, T06 to T14 (1,770)	Generated table of contents and cross-references as a template option	94%
TG8	Layout fixed to two measurements; per-measurement dates and notes	T08 (210)	Comparison device blocks for any number of measurements (3.4, working decision) and per-measurement placeholders	95%
TG7	Two templates used as a pair (appendix and summary)	T04, T05 (1,010)	A template record that defines several output documents over one block set	97%
TG11	Named signatories and their titles fixed in the template	T06 to T08, T10, T11, T13, T15 (1,000)	A device "issued by" block filled at issue from the issuing user (name, qualification, date and time, version); no fixed signatories or signature images in clinic blocks	99%
TG4	Template scripts that build the frequency deviation table and drop it when empty	T04, T09 to T11, T13, T14 (960)	A device table for frequency markers (Core, table 3.3) with a rule for absent data	100%
TG9	Per-channel values laid out as a head diagram	T10 (120)	A device per-channel head map block	100%
TG1	Clinic legends and shapes drawn over device figures	T01 to T03 (7,410)	Company-verified legend variants inside figure blocks, restricted to restating the scale ("below the norm", "near the norm mean", "above the norm"); lay headings that rename measures and the word for clinically "normal" are left out	Japanese: 94% with TG10
TG2	Japanese plus English terms in section titles	T01 to T03 (7,410)	An optional second-language term in device block titles, from a verified term list	Japanese: 100%

Not proposed: age variants of clinic text (TG3). Every age condition found sits in treatment or lifestyle recommendations, so it is left out (rule 5). Revisit only if a neutral age-dependent text appears.

Also needed, not counted in the figures (from the review):

- **Module blocks.** Each later module is a separate certified device added without changing the core (3.1). The template record needs an extension point now for module output, with the module's own version and claim boundary. The largest group is the questionnaire module: 122 questionnaire and cognitive test elements in 12 of the 16 templates, half of the 242 later-module elements; then the CP text (AI interpretation), the classifiers, the indices under licence, vigilance, the V3 sLORETA block over ICA components and spike detection (microstates are Open since 30 October).
- **Absent data.** Each device block needs a stated behaviour when its data are missing (hide, or show "not recorded"); V3 hides sections silently with block switches.
- **Anamnesis prefill.** In T04, T06, T07 and T09 to T14 the anamnesis and referral fields are filled from the patient's history form today. The history form is a questionnaire, so it is in the later questionnaire module (table 3.3). In a core report the clinician types these fields in the report editor (3.5), and the automatic prefill is lost until that module is certified: a workflow loss to discuss under I11.

Theme and layout

The I6 theme layer names fonts, colours, sizes, header and footer layout. The templates also need:

Need	Templates	Covered by the I6 theme as written	Smallest addition
Different first page (no running header on the cover)	15 of 16	no	First-page variant of header and footer
Cover: full-bleed banner bitmap (T01 to T03), full-page cover bitmap (T13), Word cover-page control with logo and text boxes, letterheads (T07, T15)	13	partly (logo)	A cover block type with a full-bleed image area and placeholders; no text inside bitmaps

Need	Templates	Covered by the I6 theme as written	Smallest addition
Running header with patient slots and page number from page 2	15	partly	Header slots for name, birth date and page number
Section numbering: typed into headings (T01 to T03, breaks when blocks are reordered) or automatic roman and arabic	13	no	Automatic heading numbering as a theme option
Start on new page, manual page breaks (25 to 76 per long template)	14	no	"Start on new page" and "keep together" as block options (mockup assumption)
Figure width options on image markers	6	no	A figure size option per device block within fixed limits
Fonts: several proprietary Japanese fonts	T01 to T03	yes, if the approved list covers Japanese	Approved Japanese fonts that embed in the PDF
Legend text coloured to match the fixed device curve colours	8	no	Legends belong to the device block; clinic colours must stay clear of the fixed z colour scale
Footnotes with citations and method notes	5	no	Footnotes in clinic text blocks, or leave out

What V4 adds to every template

A clinic template must make room for these, and the theme must not be able to hide them:

- the software and norm versions, report version and issue status (3.5, 3.8);
- the issuing clinician, date and time of issue (TG11);
- signal quality, the analysed durations and accepted trials, and any manual change of the cleaning or the segments (3.4);
- the fixed sentences from the library that describe the results and their quality (3.2); in the mockup one per marked value, the summary by fixed rule and the limits sentences (M6, M7);
- which values are marked after the correction for multiple comparisons (3.2, working decision: a false discovery rate method; how it handles correlated values is decided at implementation);
- for comparisons, the statement that earlier measurements were reprocessed (3.4);
- in the mockup, "Mockup, synthetic data" on every page.

T01 to T03 have no text at all today; these blocks would probably add one to two pages to them (estimate). A set of **mandatory device blocks** per report type is a proposed schema addition.

Leave out

Reason	Elements	Examples
Fixed verdict	79	static sentences stating a result whatever the data: no epileptiform activity, synchronised hemispheres, no significant findings, a match with a condition group, a favourable task verdict
Treatment advice	66	recommendation, medication, neurofeedback, nutrition and lifestyle texts, and their age variants
Legacy text	28	generated legacy sentences with construct labels (behaviour and figure sentences), replaced by the V4 sentence library
Control marker	27	language selectors, hidden patient-information lines, block switches
Error	20	text in the wrong language for the template, swapped labels, copies from a sibling template
Name in sentence	10	the patient's name inside generated or fixed sentences
Scanned signature	6	signature images

Claim wording

Claim wording (conditions, mental-function labels, treatment, fixed verdicts, regulatory claims, lay glosses) is flagged on every element, whatever its class: 70 of 153 clinic text elements (46%), 28 elements that need a schema addition (15 forms, 9 overlays, 4 method texts), 168 elements to leave out, 224 outside the core and the one "Core, proposed" element (T10). Many sit in long introductions and method texts that are otherwise neutral; four introductions (T06, T07, T13, T14) state a regulatory approval of the norm database. If clinic blocks stay freely editable, the core's claim boundary (ADR 0003, `intended-use.md` §4) depends on what each clinic types. I6 leaves open whether to warn on claim words in clinic blocks. Options for the product owner and the clinical lead: (a) warn on a word list in all languages; (b) block saving until the company has reviewed the text; (c) clinic texts only from company-verified variants. The findings suggest at least (b) for the first release, in which the company builds the templates anyway (3.5, working decision). A related question for the clinical lead: are reports handed to patients? The Japanese overlays are written for lay readers.

Mockup assumptions in this analysis

Also listed in `README.md` (M16 to M21). Gap ids are TG1 to TG11 ("template gap") so that they do not clash with the open question ids (G1, G2, ...) of the specification repository.

- A generated table of contents is a template option, not part of the I6 schema (TG5).
- "Start on new page", "keep together" and a figure size are block options, not in I6.
- The report language is an attribute of the template record.
- The stress classifier is counted with bio signals (dropped) where the audit shows it switched off; it has no own row in table 3.3 of 30 October, so it goes with the bio signals row. Cognitive tests are counted with questionnaires (later module, table 3.3).
- The manual ERP component tick tables are clinician forms (TG6). They could also be read as ERPC2 (open, Q7).
- Usage figures are rounded to tens so that the T codes are harder to map back to file names; anyone holding the export can still do so.